

Q22 (Convergence guarantees). Our main guarantee is conditional on convergence of the mean-field dynamics. Unconditional convergence rates for deep RF-LR remain largely open; we will clarify this and avoid implying a general unconditional guarantee. We will also point to the sharpest partial results available and clarify which assumptions are sufficient for the global-optimality conclusion given convergence.

Q23 (High-dimensional extensions of the spike mechanism). Yes in substance: the mechanism (channel dominance $\leftrightarrow$ amplified backward signal through the mixing matrix and ReLU gates) is dimension-agnostic at the ODE algebra level because it depends on the same identity

$$H_2(c_2; x, W) = \sum_{j=1}^r W_{c_2,j} f_j(x; W)$$

and the channelwise backprop objects that feed the mean-field equations. What we do not currently have for general input dimension d is a clean theorem isolating well-separated (non-overlapping) spikes in feature space; overlapping channel contributions when $d > 1$ complicate bookkeeping and the rigorous two-point caricature used in Theorem 4.1. We will state this distinction clearly in the camera-ready and add empirical evidence in higher d where space permits.

Additional context. The explanatory content is not the simplified two-point, two-channel state-ment itself, but the positive feedback it isolates. Fix a second-layer neuron index c_2 and recall $H_2(c_2; x, W) = \sum_{j=1}^r W_{c_2,j} f_j(x; W)$. For ReLU, gating enters mean-field gradients through $\varphi'_2(H_2) = \mathbf{1}\{H_2 > 0\}$. The backward signal for channel k (cf. the ODE for $\partial_t w_1(\cdot, k)$) is built from expectations over C_2 of terms that couple (i) the mixing weight $W_{C_2,k}$, (ii) the gate $\mathbf{1}\{H_2(C_2; x, W) > 0\}$, and (iii) upstream factors (output error and later-layer weights). When channel k dominates, neurons c_2 that contribute to the drift along channel k are precisely those with the gate “on.” Across c_2 , the effective weights in the expectation are then positively aligned with $W_{c_2,k}$ and with $\mathbf{1}\{H_2(c_2) > 0\}$, producing a large contribution to the drift of the k th channel, which in turn pushes f_k further in the same direction: a reinforcing loop. The ODE-level mechanism is dimension-agnostic; the two-point theorem is a minimal tractable instance.

Visualization note. The figures emphasize $d = 1$ mainly for clarity of visualization; the same feedback loop operates when $x \in \mathbb{R}^d$. Overlapping spikes mainly affect how we illustrate and formally isolate regimes, not the core mean-field structure above.

Q24 (Sensitivity to the frozen feature draw). The analytical conditions are not tied to one sampling recipe. Once mixing matrices are frozen, the arguments use an almost-sure channel-aggregation bound (schematically $\|W^{(\ell)}\|_{\infty,1} \leq rK$) and a richness/diversity assumption for the frozen first-layer features. In principle each frozen $W^{(\ell)}$ can be drawn from any law with finite support (or any law satisfying the same $\ell_{\infty,1}$ control almost surely); there is no Gaussian-specific requirement. In experiments we often use unbounded Gaussian initializations for simplicity at finite width; that is an engineering choice while the theorems use stability/richness predicates. Nonnegative / NMF-style $W \geq 0$ matches the same boundedness template and is intuition for some collapse modes; the ReLU dead-gate caveat is separate (high probability in r only as a heuristic; cf. Q10). At finite width, performance can vary with the draw. We have not run a systematic post-review sweep over independent frozen-feature draws with optimization held fixed; we will state this in the camera-ready, separate proved assumptions from mean-field idealization, and add compact multi-draw diagnostics if space permits or list them as future work.